



Absolute Value and Exponential Functions Quiz 1

Form A

Answer key + scoring guidance

DIRECTIONS

10 items • 12 points • 25 minutes • Form A

Choose one answer unless an item says Select all. Show work inside each item panel. Label graphs, endpoints, units, and constraints. Use a calculator only where the item says CALCULATOR AVAILABLE. No reference sheet is provided.

1

SOLVE • COORDINATE GRAPH • CALCULATOR AVAILABLE

1 PT

State the range of $f(x) = 2|x + 5| + 1$.

- ☐ A $y \leq 1$
- ☐ B $y \geq -5$
- ☐ C $y \geq 1$
- ☐ D all real numbers

Answer (1 pt): $y \geq 1$ **Solution:** The graph opens upward from the vertex $(-5, 1)$, so 1 is the minimum output.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.; Confuses input uniqueness with repeated outputs or reads $f(x)$ as multiplication.

2

SOLVE • SYMBOLIC • CALCULATOR AVAILABLE

1 PT

Given $f(x) = -|x - 5| + 5$, find $f(8)$.**Answer (1 pt):** 2**Solution:** $|8 - (5)| = 3$, so $f(8) = -1(3) + 5 = 2$.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.; Confuses input uniqueness with repeated outputs or reads $f(x)$ as multiplication.

3

SOLVE • COORDINATE GRAPH • NO CALCULATOR

1 PT

State the range of $f(x) = -|x - 1| + 5$.

- ☐ A all real numbers
- ☐ B $y \geq 5$
- ☐ C $y \geq 1$
- ☐ D $y \leq 5$

Answer (1 pt): $y \leq 5$ **Solution:** The graph opens downward from the vertex (1, 5), so 5 is the maximum output.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.; Confuses input uniqueness with repeated outputs or reads $f(x)$ as multiplication.

4

CONNECT • COORDINATE GRAPH • NO CALCULATOR

1 PT

State the vertex and opening direction of $f(x) = 2|x - 4| + 2$. Select all statements that are true.

- ☐ A The result is not vertex (0, 2); opens up.
- ☐ B The result is vertex (4, -2); opens down.
- ☐ C The result is vertex (4, 2); opens up.
- ☐ D The result is vertex (-4, 2); opens up.

Answer (1 pt): The result is vertex (4, 2); opens up.; The result is not vertex (0, 2); opens up.**Solution:** Absolute-value vertex form reveals vertex (4, 2); the sign of 2 gives the opening direction.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.; Confuses input uniqueness with repeated outputs or reads $f(x)$ as multiplication.

5

CONNECT • MULTIPLE REPRESENTATION • NO CALCULATOR

1 PT

Match each function to its vertex. Each one uses key features.

1.	An absolute-value function has $f(0) = 4$, $f(1) = 5$, and $f(2) = 4$. Give the vertex.	Match: _____
2.	Given $f(x) = 2 x + 4 + 1$, find $f(-2)$.	Match: _____
3.	State the vertex and opening direction of $f(x) = 2 x + 1 - 3$.	Match: _____

Answer bank: A. (1, 5) • B. 5 • C. vertex (-1, -3); opens up

Answer (1 pt): 1 → (1, 5); 2 → 5; 3 → vertex (-1, -3); opens up**Solution:** 1: Equal outputs at $x = 0$ and $x = 2$ sit symmetrically about $x = 1$, so the vertex is (1, 5). | 2: $|-2 - (-4)| = 2$, so $f(-2) = 2(2) + 1 = 5$. | 3: Absolute-value vertex form reveals vertex (-1, -3); the sign of 2 gives the opening direction.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.; Confuses input uniqueness with repeated outputs or reads $f(x)$ as multiplication.

6

SOLVE • SYMBOLIC • CALCULATOR AVAILABLE

1 PT

Factor $x^2 - 9$.

Answer (1 pt): $(x - 3)(x + 3)$ **Solution:** Use $a^2 - b^2 = (a - b)(a + b)$: $(x - 3)(x + 3)$.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Finds numbers with the correct sum but incorrect product, or omits a greatest common factor.

7

SOLVE • COORDINATE GRAPH • CALCULATOR AVAILABLE

1 PT

State the vertex and opening direction of $f(x) = 2|x + 2| + 1$.

- ☐ A vertex $(-2, 1)$; opens up
- ☐ B vertex $(0, 1)$; opens up
- ☐ C vertex $(-2, -1)$; opens down
- ☐ D vertex $(2, 1)$; opens up

Answer (1 pt): vertex $(-2, 1)$; opens up**Solution:** Absolute-value vertex form reveals vertex $(-2, 1)$; the sign of 2 gives the opening direction.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.

8

APPLY + EXPLAIN • TABLE • CALCULATOR AVAILABLE

2 PTS

An absolute-value function has $f(-2) = -2$, $f(-1) = -4$, and $f(0) = -2$. Give the vertex. Show enough reasoning to justify your answer.**Answer (2 pts):** $(-1, -4)$ **Solution:** Equal outputs at $x = -2$ and $x = 0$ sit symmetrically about $x = -1$, so the vertex is $(-1, -4)$.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.

9

CHECK + JUSTIFY • TABLE • NO CALCULATOR

1 PT

A quadratic has $f(-6) = 8$, $f(-5) = 7$, and $f(-4) = 8$. Give the vertex. A student reported $(-6, 8)$. Identify the likely error and give the correct result.**Answer (1 pt):** Correct result: $(-5, 7)$.**Solution:** The reported result is inconsistent with the required operation or representation. The equal outputs at $x = -6$ and $x = -4$ sit symmetrically about $x = -5$, so the vertex is $(-5, 7)$.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Confuses zeros, x-intercepts, vertex coordinates, and the value of a function.

10

APPLY + EXPLAIN • COORDINATE GRAPH • NO CALCULATOR

2 PTS

Apply the lesson strategy to this unfamiliar version. State the range of $f(x) = 2|x| - 2$. State the result and justify why it is reasonable.

Answer (2 pts): $y \geq -2$ **Solution:** The graph opens upward from the vertex (0, -2), so -2 is the minimum output.**Scoring:** Award full credit for the correct response and required reasoning; use the teacher solution for acceptable equivalents.**Watch for:** Treats absolute value as one case instead of a distance with two possible directions.; Confuses input uniqueness with repeated outputs or reads $f(x)$ as multiplication.